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Diel movements and flight altitudes of the European storm-petrel in the North Atlantic and their implications for conservation

Clara Morey[1]✉, Mikel Yarza[3], Juan Bécares[1], Jabi Zabala[2], Marcel Gil-Velasco[1], Laura Zubiate[4], Marc Illa[1], Iker Apraiz[2], Saúl Román[1] & Iñigo Zuberogoitia[3]

Understanding the movement patterns and foraging areas of seabirds is critical for their conservation and for mitigating potential impacts from human activities, such as offshore wind farm (OWF) development. We provide the first GPS data on foraging trips and travel altitudes of European storm-petrels (ESP) *Hydrobates pelagicus* in the Bay of Biscay, near planned OWF development areas. We obtained 28 complete trips from 19 individuals during the incubation periods of the 2023 and 2024 breeding seasons at three colonies along the Spanish Cantabrian shore. Distribution models for ESP were developed using the maximum entropy (MaxEnt) method for each colony, and flight heights were measured using GPS triangulation, analysing data separately for daytime and nighttime. Mean trip duration was 2.7 days and mean trip distance travelled was 803 km, with no relevant differences among colonies. Our results indicate that foraging areas were concentrated along the edges of both the French and Iberian continental shelves. However, at night, ESP showed a more concentrated use of the continental shelf edge and exhibited a more coastal distribution, occasionally foraging in bays and estuaries and showing inter-colonies nocturnal movements. This nocturnal scenario may pose a risk of direct impact of OWFs on the breeding population of the species in the Gulf of Biscay.

Keywords GPS tracking, Seabird distribution, Offshore wind farms, Flight heights, Foraging areas, Gulf of Biscay

Numerous offshore wind farms (OWFs) have been planned, approved, and are under construction worldwide to meet the increasing demand for renewable energy resources[1]. The OWF development has the potential to strongly affect the seabird distribution and habitat use due to different behavioural reactions, ranging from complete avoidance leading to displacement from a former habitat, to attraction causing an increasing risk of collision with the turbines[2,3]. Unfortunately, there is not a positive relationship between the increase of knowledge of behavioural, ecological and population effects of OWFs on local, migrant and wintering seabird species and the development and installations of new projects[4–6].

Spanish waters are very deep due to the narrow width of its continental shelf. Floating OWFs can be located further offshore, where they can benefit from stronger winds and remain out of sight[7]. Therefore, floating structures could become key to the development of offshore wind energy in Spain. The Spanish Government approved the Maritime Spatial Plans on 28 February 2023[7], identifying high-suitability zones for the future deployment of commercial offshore wind energy infrastructure. Eight of these Offshore Renewable Energy High Potential Zones are located within the North Atlantic marine demarcation. These areas, labelled NOR1 to NOR8, cover a total surface area of 2688.61 km^2 and are located between 14.5 and 31.5 km from the coast, offshore of the autonomous communities of Galicia and Asturias in northern Spain (Fig. 1). They are situated north of and directly adjacent to the recently designated Special Protection Area for Birds 'Corredor migratorio galaico-cantábrico occidental ES0000554', as published on 30 December 2023 in Document BOE-A-2023-26741. Additionally, the Maritime Spatial Plans also include proposals for potential future areas dedicated to research, development and innovation in renewable energy.

[1]CORY'S – Investigación y Conservación de la Biodiversidad, Calle Maladeta, 22. ENT 2ª, 08016 Barcelona, Spain. [2]Department of Zoology and Animal Cell Biology, University of the Basque Country UPV/EHU, C/Paseo de la Universidad 7, 01006 Vitoria-Gasteiz, Álava-Araba, Basque Country, Spain. [3]Estudios Medioambientales Icarus S.L., C/San Vicente, 8. 6 Planta, Dpto 8. Edificio Albia I., 48001 Bilbao, Spain. [4]BiMEP, Atalaia 2, bajo, 48620 Armintza, Bizkaia, Spain. ✉email: cmorey@corys.es

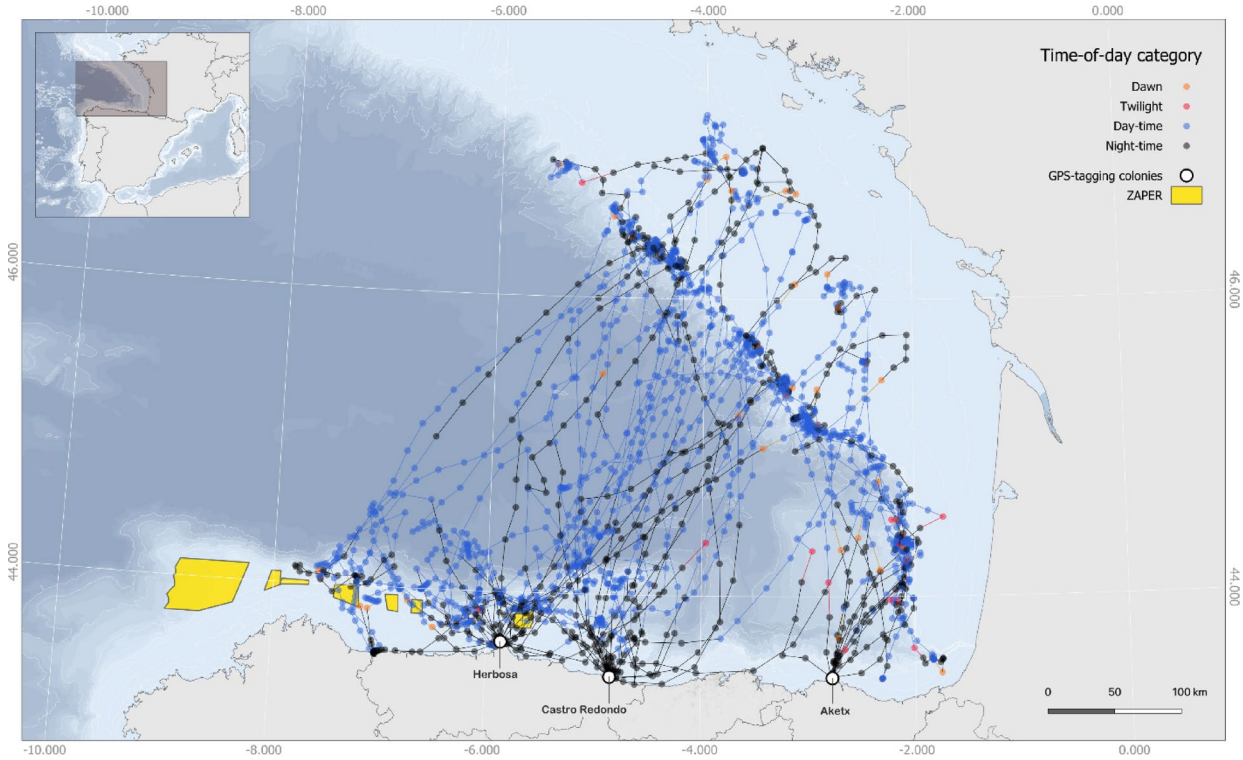


Fig. 1. Foraging tracks of 19 European storm-petrels breeding on Aketx islet (Basque Country) and on Herbosa and Castro Redondo islets (Asturias), and areas with high potential for offshore wind development (ZAPER). GPS data from complete trajectories recorded in 2023 and 2024 are categorised by time of day. Incomplete journeys and trips shorter than 3 h were excluded from the analyses. Map was created using QGIS version 3.40.12- 'Bratislava'.

The European storm-petrel *Hydrobates pelagicus* is a pelagic seabird that only comes ashore to breed or to find a breeding partner and/or a suitable nesting site[8]. In the Spanish North Atlantic demarcation, approximately 1700 pairs are thought to breed, including Basque Country: 373 pairs; Cantabria: 390 pairs; Asturias: 750 pairs; and Galicia: 186 pairs[9]. The planned increase in the number of OWFs poses a threat to the species, as it might result in either a displacement from foraging areas, direct collisions with turbines and/or other related impacts. Overall, the potential impacts of OWFs on small petrels are poorly understood[5], since at-sea distribution during the breeding season, and particularly the non-breeding season, is insufficiently known for many species, and consequently so is our ability to assess spatial risk from renewable energy sectors[5]. Formal assessments conclude that collision risks are low for petrels, partly because their typical flight heights are below the arc of most turbines[5]. However, the impact could be more significant in certain species, such as storm-petrels, since they could be attracted by the artificial lights of offshore platforms[3]. Indeed, the European storm-petrel, the smallest palearctic seabird, ranks among the species with the highest sensitivity index to offshore wind farms, out of the 30 seabird species considered[10] and comes in 18th position out of the 47 species assessed in Spanish waters[11]. However, the lack of empirical data only allows us to hypothesize the effects of these structures on these seabirds. Until now, foraging ranges of storm-petrels in the area have been defined from vessel-based transect surveys[12,13], and the densities derived from them are known to be extremely low[14,15]. Vessel-based observations and those conducted at OWFs sites suggest European storm-petrels generally fly within 2 m of the sea surface, but occasionally up to heights that would put them at risk of collision with wind turbine blades[6,16,17]. Nevertheless, nocturnal flight behaviour remains completely unknown. Ringing programs developed on the top of cliffs and islands evidence that European storm-petrels prospect elevated points during their night flights[9].

However, in recent years, the increasing miniaturisation and affordability of GPS devices have led to an upsurge in seabird tracking, which has enabled a colony-focused approach to mapping seabird marine distribution[18]. Although these tiny tracking devices cannot provide the finest scale resolution and are battery-constrained, they can still track entire foraging trips (diurnal and nocturnal) that serve to describe the species' foraging ecology with significant detail[3,19]. This has allowed us to identify key offshore areas for European storm-petrels[3,19–21], marking a significant step forward in the species' at-sea conservation and management.

This study provides the first report of interannual foraging movements in European storm-petrels breeding at three Cantabrian colonies in the Gulf of Biscay during the incubation period. We provide track statistics for European storm-petrels from the different breeding colonies and seasons studied, and we measured flight heights using GPS triangulation, subsequently determining the altitude error. We developed European storm-petrel distribution models using the maximum entropy method and incorporated both static and dynamic oceanographic explanatory variables across the Gulf of Biscay. All analyses were conducted separately for

Breeding colony	Season	GPS deployed	GPS recovered	Positions recovered	Total trips recovered	Round trips
Aketx	2023	20	5	516	9	6
Herbosa	2023	15	5	528	10	8
Herbosa	2024	6	3	327	5	4
Castro Redondo	2024	9	6	729	12	10

Table 1. Number of European storm-petrel individuals Hydrobates pelagicus tagged by breeding colony. Excluding incomplete journeys and trips shorter than 3 h.

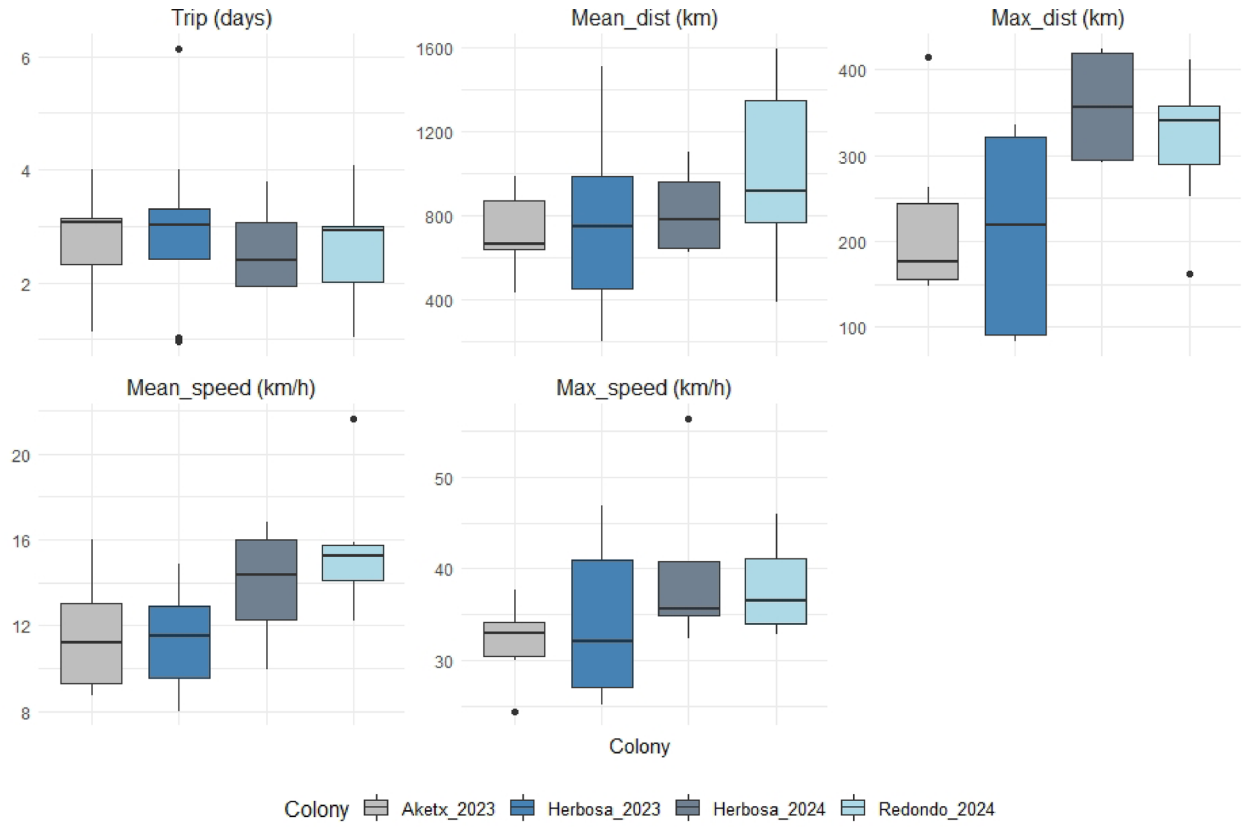


Fig. 2. Boxplots showing the median, interquartile range (IQR), and outliers beyond 1.5 times the IQR of foraging track statistics of European storm-petrels breeding on Aketx Islet (Basque Country), Herbosa, and Castro Redondo islets (Asturias), from complete trajectories.

daytime and nighttime data to compare different behaviours. Using these spatial data, we examine and discuss the diurnal and nocturnal marine distribution of European storm-petrels in the Gulf of Biscay and its relation to significant marine threats, such as offshore wind farms.

Results

Foraging trip characteristics

Overall, we obtained 28 round trips (foraging trips) from 19 individuals (Table 1, Fig. 1 & Supplementary Material Figs. S1–S5, Tables S1–S4). Recapture rates varied according to breeding colony and season: 0.25 for Aketx colony (2023); 0.33 in 2023 and 0.42 in 2024 for the Herbosa colony; and 0.66 for the Castro Redondo (2024).

A total of 1363 day positions and 733 night positions were recovered from 28 complete and 8 incomplete trips. Considering complete trips only (Fig. 2, Supplementary Material Table S5), the mean trip duration was 2.7 days (66 h; range = 23 h to 6.1 days) and the mean trip distance travelled was 803 km (range = 200–1590 km). There were no differences among colonies for duration ($F = 0.16$; $P = 0.95$) or distance travelled ($F = 1.18$; $P = 0.34$; Fig. 2). The average maximum distance to the colony was 278 km (range = 90–423 km), but there were significant differences among colonies, since, in 2024, those from Castro Redondo and Herbosa reached larger distances than those from Aketx in 2023 ($F = 19.52$, $P < 0.001$; Fig. 2). To reach such distances, instead of increasing the duration of the trips, in 2024, the storm-petrels from Castro Redondo and Herbosa flew significantly faster on average than those from Herbosa and Aketx in 2023 ($F = 99.72$; $P < 0.001$). In contrast, the average maximum

speed reached was 36 km/h (range 24.4–56.4 km/h) and did not differ statistically among colonies ($F = 1.487$, $P = 0.24$). Detailed trip characteristics for each individual from each colony are provided in Supplementary Tables S1–S4 and Figs. S2–S5.

Habitat modelling with GPS data

The distribution models show foraging areas following the edge of both the French and the Iberian continental shelves (Fig. 3). The relative importance of individual explanatory variables included in the model was evaluated using a Jackknife test, with contributions measured by the AUC (Area Under the Curve); in this way, distance to the breeding colony (65%), bathymetry (15%), and sea surface temperature (9%) were identified as the most important variables for European storm-petrels in the combined model within these areas (Supplementary Material Table S6). At night, the species exhibited a much more coastal distribution, with more intense and concentrated use of the continental shelf compared to daytime activity (Fig. 3), and even occasionally foraging on bays and estuaries (Supplementary Material Fig. S3).

Flight altitude

Results of the calibration process indicated that during feeding or resting behaviour, storm-petrels flew at an average altitude of less than 1.5 m above sea level (daytime: mean = −0.29 m, IQR = 12.79; nighttime: mean = 1.04 m, IQR = 10.0). During movement, they generally flew below 5 m on average (daytime: mean = 3.22 m, IQR = 11.32; nighttime: mean = 2.28 m, IQR = 10.66). However, storm-petrels occasionally reached higher altitudes, exceeding 20 m in 2.0% of daytime movements (n = 589) and 0.9% of nighttime movements (n = 353). Only five locations (0.9%) during daytime movements falling within the rotor swept zone (> 25 m) and only one fix during nighttime (0.3%) (Fig. 4).

Flight altitude did not differ significantly between day and night ($F = 0.884$, $P = 0.35$). In contrast, behaviour had a significant effect ($F = 16.579$, $P < 0.001$), with higher flight heights during movements than during resting or feeding. The interaction between time of day and behaviour showed a non-significant trend ($F = 2.813$, $P = 0.09$).

Discussion

The rapid increase in potential hazards for seabirds urges the undertaking of further research in the short term, with the help of modern technologies. Nevertheless, in most cases, new impacting structures are developed before fully understanding the true effects on the target species. In our case, foraging movements of European storm-petrels in the Gulf of Biscay have been barely described, principally from vessel-based transect surveys[15]. This survey methodology is also used to analyse the spatial use and flight behaviour of seabirds in

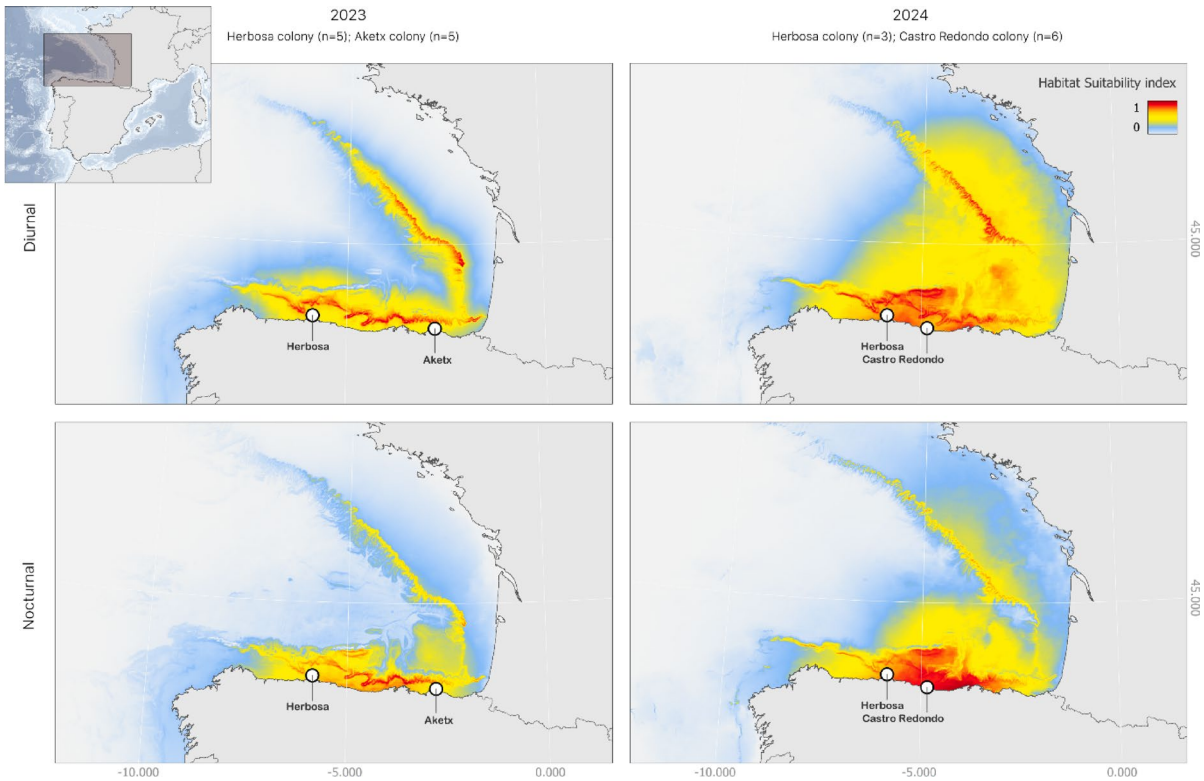


Fig. 3. European storm-petrel distribution models (MaxEnt) from the Aketx, Castro Redondo, and Herbosa colonies generated for the Bay of Biscay. (**a**) 2023 diurnal; (**b**) 2024 diurnal; (**c**) 2023 nocturnal; (**d**) 2024 nocturnal. GPS data projected onto adjacent storm-petrel colonies and weighted according to colony size (number of breeding pairs). Map was created using QGIS version 3.40.12- 'Bratislava'.

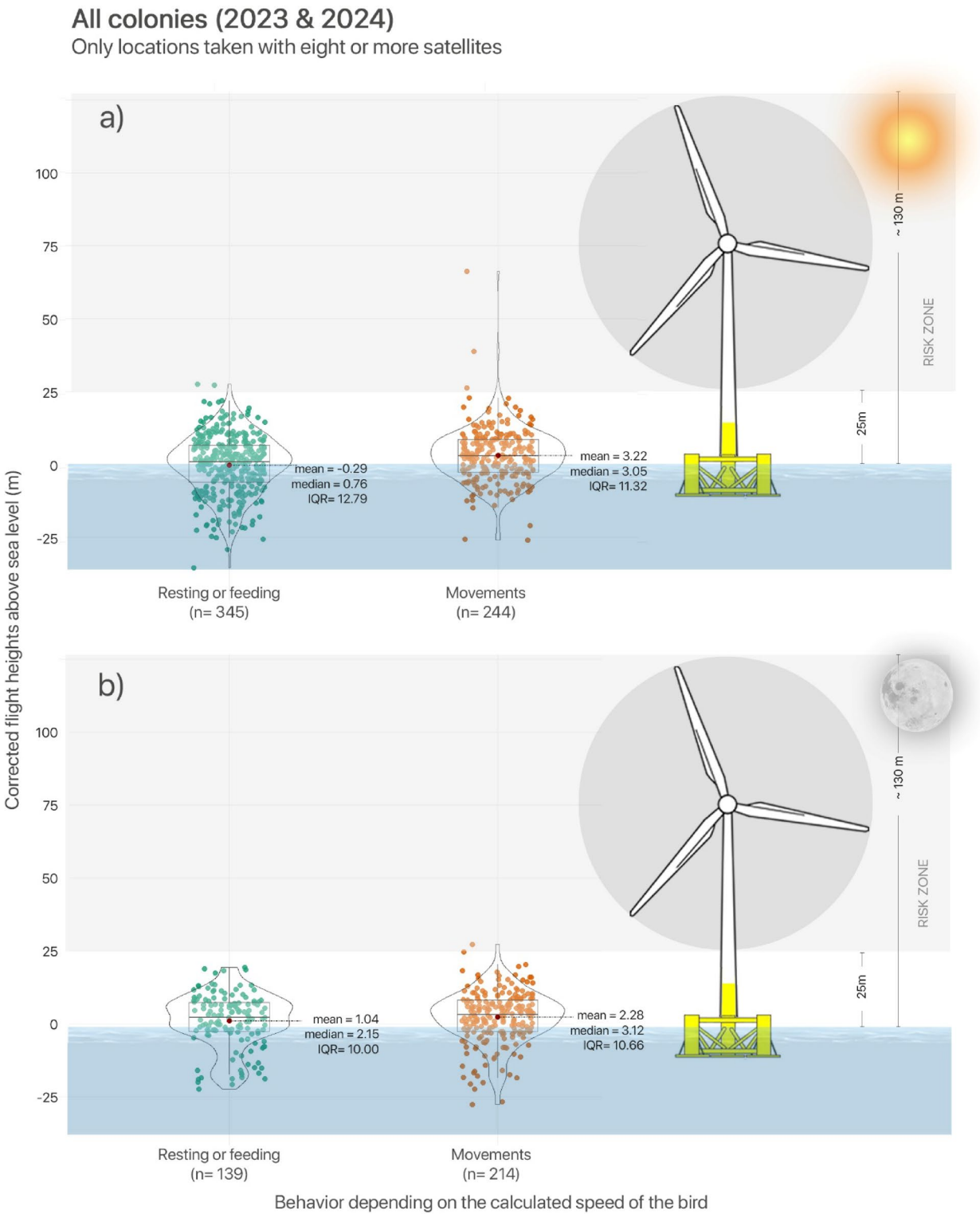


Fig. 4. Daytime (**a**) and nighttime (**b**) flight heights based on the behaviour of European storm-petrels tagged on Herbosa, Castro Redondo and Aketx islets in 2023 and 2024 (only data recorded with 8 or more satellites are included). A scale model of a BiMEP wind turbine is shown for comparison.

Environmental Impact Assessments for OWFs[22]. However, the low detectability of the species in coastal waters during daylight[15] was not correlated with the size of the breeding colonies[23]. European storm-petrels return to the breeding colonies during the hours of darkness, and hence the pattern of marine distribution is likely to exhibit pronounced variations from day to night[5]. We provide here the first GPS data of foraging movements of

European storm-petrels in the Bay of Biscay and nocturnal trips from the foraging areas to the colonies, where OWFs are planned to be installed.

Our data showed that European storm-petrels actively forage through the continental shelf edge, where strong upwelling of nutrient-rich cold-water results in high food availability for the species in the surface layers[3]. The maximum straight-line distance reached from the Cantabrian colonies was similar to those reported for both the Atlantic and the Mediterranean sub-species of the European storm-petrel[18,20,24]. Most of the tagged individuals travelled directly from the breeding colony to the northern part of the Bay of Biscay, to the areas above the submarine canyons of the Armorican shelf and Aquitaine shelf. In fact, European storm-petrels flew at maximum speed during the night, and once they reached the foraging area, they conducted prospecting flights and remained in the area on average three days and two nights. These results are consistent with previous findings showing that European storm-petrels exhibit a considerably more pelagic distribution during daylight hours compared to night, reflecting offshore foraging during the day and returning to the colony at night during darkness[3,20]. Actually, the southern part of the Bay of Biscay was used mainly during the night as transit areas, but some individuals also foraged there, in the southern shelf edge, and one individual entered the estuary of Eo river protected by the darkness. According to previous studies conducted in the Bay of Biscay, European storm-petrels forage on littoral and intertidal organisms (gobies and isopods) at night in estuaries near breeding colonies, particularly on water fleas (*Eurydice pulchra*)[25,26]. In fact, as our results and others showed, European storm-petrels avoid coastal waters during the day, presumably to prevent attacks of avian predators, as for example Yellow-legged gulls *Larus michahellis*[27,28] and Peregrine falcons *Falco peregrinus*[29].

Our results also show that, although the duration and distance travelled were similar across the four colony-year combinations, storm-petrels from Castro and Herbosa in 2024 travelled further to reach foraging areas and increased their mean speed, presumably to cover more distance in the same amount of time. Moreover, as previously shown for Mediterranean storm-petrels, suitable foraging areas vary spatially and temporally[21]. This may be related to the changing dynamics of the nutrient balance in the Bay of Biscay, in accordance with findings indicating that the probability of foraging increased towards cold, nutrient-rich and shallow waters with upwelling dynamics[21]. However, individuals appear much more site-faithful compared to other colonies, particularly along the edge of the French continental shelf. This strong fidelity has important conservation implications, as it likely indicates lower flexibility to buffer disturbance in foraging areas.

According to previous studies[3], vessel-based visual surveys, necessarily conducted during daylight and calm sea states[22], may not fully represent their marine distribution integrated across the entire diurnal cycle and in all weather conditions. This is particularly true for our study area, where previous studies[15] barely obtained a few records of the species during vessel-based surveys. They suggested that European storm-petrels were more likely to be found at a distance between 20 and 30 km from the coast, which corresponds with the continental shelf edge in most of the surveyed area. However, they failed to detect them in the neighbouring waters of the main colonies, since birds were almost entirely absent from this area during daylight. Ringing programs developed during the last three decades through the islets and coastal cliffs of the Bay of Biscay, probed inter-colonies nocturnal movements, not only assigned to transients but also to breeders[23], supported by the observation of a tracked breeding individual from Castro Redondo islet at Herbosa islet (Supplementary Material Fig. S5). Actually, the nocturnal scenario in the continental shelf, where thousands of European storm-petrels (transient and breeders) normally fly, is totally different to the diurnal one. In fact, our data proved that European storm-petrels used the continental shelf not only to move from the colony to the foraging places, but also to forage, and the potential overlap between foraging areas and high-suitability zones for the future deployment of OWFs within the North Atlantic demarcation is especially significant for the colonies in Asturias and Galicia, where all such zones are located.

Although there is no information in the literature regarding the extent of avoidance behaviour, displacement, and barrier effects of OWFs on European storm-petrels[6], the species travels long distances to target specific oceanographic features[21] and displacement from these important foraging areas would likely have negative consequences. Additionally, it is essential to carry out remote tracking studies of the European storm-petrel in Galicia—where several high-suitability zones for the future deployment of OWFs are being established—as none have been conducted there to date.

Flight altitude is considered the most important factor when estimating collision risk for marine bird species at OWFs[17]. Vessel-based observations indicate that European storm-petrels generally fly within 2 m of the sea surface, although occasional flights up to 5 m have been reported[16]. Consistent with these observations, our data show that European storm-petrels predominantly flew at low altitudes, within 0–4 m above sea level. Although storm-petrels may occasionally fly at higher altitudes, the large uncertainty associated with our altitude estimates (semi-IQR = 9.03 m; IQR = 18.05 m) limits our ability to draw robust conclusions regarding fine-scale vertical use. Nevertheless, a proportion of movement fixes indicated altitudes exceeding 10 m (18.2% and 16.1% of daytime and night-time movement fixes, respectively), and a very small fraction of locations (< 1% of fixes) were estimated to fall within the rotor-swept zone. We consider these latter cases most likely to reflect measurement error, with true flight heights remaining below the collision risk zone (Fig. 4), consistent with other research reporting vertical measurement errors of ~ 20 m for GPS and barometric altimeters that can artificially broaden height distributions[30].

In contrast, in the vicinity of the breeding colonies, prospecting petrels regularly fly over islets and cliffs where they are ringed[23] (pers. observ.), with most nests in the monitored colonies located at elevations exceeding 15 m. This raises concern about the potential attraction of petrels to offshore platforms situated close to breeding colonies, which could be explored as alternative nesting sites, similarly to what has been documented for small islets. A key factor in this process is the extent to which nocturnally active seabirds, such as shearwaters and storm-petrels, may be attracted to the artificial lighting required for turbine operation[6,31,32].

Light attraction and disorientation are a very well-documented behaviour of petrels across the world[5]. Attraction per se (i.e. "phototaxis") could potentially operate over ranges of tens of km and may result in birds being displaced from foraging areas and activities[6] and may affect the number of birds brought into the vicinity of wind turbines and associated structures, vessels and shore facilities[33]. Light-induced disorientation alters the flight paths of birds when in close proximity (i.e. within tens of metres), and may cause birds to circle light sources for many hours[34], affecting the length of time birds remain within the proximity of potential collision surfaces, with obvious implications for collision risk[33]. As suggested by other authors[6], the impact of light attraction on flight height must be considered, as it remains unclear to what extent such attraction or disorientation may lead birds flying close to the sea surface—below rotor-swept height—to ascend into the rotor-swept zone. Nevertheless, this seems likely in the case of storm-petrels stranded on oil platforms. Although the clear effect has not yet been assessed in the case of storm-petrels[3,35], previous research showed that, in Atlantic Canada, marine offshore lights of oil and gas platforms and similar, attracted birds and that the Leach's storm-petrel *Hydrobates leucorhous* was the species most frequently found stranded[36].

In conclusion, based on the flight altitudes estimated for European storm-petrels in the Bay of Biscay, our results suggest a low collision risk associated with offshore wind farms (OWFs), at least during the breeding season. However, our data also show that European storm-petrels use the Spanish continental shelf not only as a commuting corridor between breeding colonies and foraging grounds, but also as an important foraging habitat. The markedly more coastal nocturnal distribution of the species, compared to the daytime patterns, highlights the potential for spatial overlap between key foraging areas and zones identified as high suitable for the future OWFs deployment, which should therefore be carefully considered.

In addition, the potential hazard posed by floating platforms warrants further attention, as these structures may act as attraction or prospecting sites for storm-petrels. Mitigation measures aimed at reducing light attraction, such as the use of less attractive wavelengths, alternative colours, or light shielding, should be explored. Further research is also needed to assess avoidance behaviour, displacement, and barrier effects of OWFs on European storm-petrels. As the effectiveness of mitigation strategies is likely to vary among species, weather conditions, and seasons[35], long-term monitoring of the main breeding colonies in the eastern Bay of Biscay[9,37] will be essential to evaluate population trends and to assess the cumulative impacts of OWFs over time[3].

Methods

Study areas

The study was carried out on three European storm-petrel colonies along the Spanish Cantabrian Sea: Aketx Islet (43.4511, −2.7753) (Basque Country), one of the most important colonies of the species in the Bay of Biscay; Herbosa Island (43.6632, −5.8637) and Castro Redondo Islet (43.4463, −4.8453), both located off the central and eastern coast of Asturias (Fig. 1 and Supplementary Material Fig. S6.

Aketx is a small limestone islet of 6 ha, 80 m high, situated 800 m offshore to the north of the coast. The eastern part of the island has vertical, inaccessible cliffs with crevices and caves, where European storm-petrels breed. The European storm-petrel population for Aketx islet is estimated at around 250 breeding pairs[9].

Herbosa is a small islet of 4.2 ha situated 700 m off the northernmost cape of Asturias (Cabo Peñas) and it is surrounded by reefs and small islets. The Herbosa storm-petrel colony is quite isolated from others known colonies of the species, since it's around 130 km apart from the nearest colony to the east (Os Farallons, Galicia) and 60 km from the nearest colony to the west (Islote Gaviota, Asturias). The storm-petrel population in this place is estimated between 32 and 40 breeding pairs (2024 census, own unpublished data). Both Aketx and Herbosa islets host breeding colonies of Yellow-legged Gulls *Larus michahellis* and European Atlantic Shags *Gulosus aristotelis*.

Castro Redondo (43.4463, −4.8453) is a small islet of 0.23 ha, situated only 60 m off the Eastern Asturian coast (Llanes). This colony is close to other storm-petrel colonies on other islets along the Llanes coast, so, despite Castro Redondo might host only 45–50 breeding pairs (2024 census, own unpublished data), around 750 breeding pairs are estimated for the Llanes coast, which represents the main species stronghold in the entire Iberian Cantabrian region[9].

Sampling protocol

In 2023, we surveyed the Aketx and Herbosa European storm-petrel colonies, inspecting and checking every hole, crevice, and nest box in the surrounding accessible areas to locate isolated nests that were difficult to find. In 2024, the survey was repeated at the Herbosa colony, and the Castro Redondo colony was monitored for the first time.

At the Aketx colony, from 1993 to 2014, we conducted systematic surveys, consisting of a ringing and reproductive monitoring program[37]. We systematically monitored the breeding success in 17 nests situated in crevices on the western side of the islet. There were no previous monitoring projects in the Asturias islets (Herbosa and Castro Redondo), although 40 nest boxes were set up at the Herbosa colony, approximately 20 years ago.

Adults were captured by hand at the nest, and in the more inaccessible nests, we used a large stick with a loop at the end to capture storm-petrels, in an effort to minimize stress.

We checked every nest weekly and tried to capture incubating adults in order to set a GPS device. We used PathTrack (Otley, UK) nanoFIX® GEO-MINI GPS loggers (ca. 0.9 g, < 4% bird body mass), set to record one fix per hour during incubation. Tag schedules were designed to maximise temporal resolution, while ensuring that tags remained active until the completion of at least one foraging trip. The tags were attached to the central four tail feathers with two narrow strips of Tesa® tape, resulting in a total load of 1.1 g[20] (Supplementary Material Fig. S6).

Breeding colony	Season	GPS devices deployed	Deployment dates
Aketx	2023	20	June 9th and 26th, and July 10th and 13th, 2023
Herbosa	2023	15	July12th, 2023
Herbosa	2024	6	July 1st and 2nd in 2024
Castro Redondo	2024	9	June 25th and 26th, 2024

Table 2. Number of European storm-petrel individuals Hydrobates pelagicus tagged by breeding colony in the Bay of Biscay.

In total, we deployed 50 GPS devices on storm-petrels. Dates of deployment are as outlined in Table 2. All GPS devices were deployed and recovered during incubation period.

We excluded all incompletely observed trips. Multiple trips per deployment could occur if the bird was not recaptured during the first return to the colony. We excluded GPS positions located within the breeding colony. We considered only complete tracks: 6 tracks for Aketx colony (2023), 8 and 4 tracks respectively for Herbosa in 2023 and 2024, and 10 tracks for Castro Redondo (2024). We measured trip duration (hours and days), total distance travelled (km), maximum straight-line distance from the colony (km), mean travel speed (km/h) and maximum travel speed (km/h).

Behaviours at sea were distinguished using the EMbC algorithm[38], which classified behaviours based on speed and turning angle: resting (low speed, low turning angle), intensive foraging (low speed, high turning angle), extensive foraging (high speed, high turning angle), and traveling (high speed, low turning angle. However, given the 1-h GPS interval, the classifications should be considered approximate. All analyses were performed by the authors using R version 4.0.5 for Windows (R Core Team 2021; https://www.r-project.org).

Habitat modelling with GPS data

We developed European storm-petrel distribution models (SDM) using the maximum entropy method (MaxEnt software, version 3.4.4)[39], based on the storm-petrel's tracking data obtained in three different colonies during the breeding season of 2023 and 2024 in the Bay of Biscay. In order to work out potential ecological differences, we split the locations into daytime (6AM to 9:59PM, local time) and nighttime (10PM to 5:59AM, local time) and generated different models accordingly.

Modelling scope and selection of environmental variables

The scope of the study was limited to the Bay of Biscay. All the variables used had a spatial resolution of 0.01 degrees per pixel and shared the same coordinate system (datum WGS-1984). Seabed slope, bathymetry, sea surface temperature (SST), chlorophyll-a concentration, and distances to colonies were selected as they reflect key oceanographic conditions influencing seabird distribution and spatial patterns[40,41]. Static variables included bathymetry, slope, distance to the breeding colony and distance to the 200 m isobath, derived from the Emodnet model[42]. The dynamic variables included chlorophyll-a concentration and sea surface temperature, corresponding to July 2023 and July 2024, respectively[43], with each variable taken for the specific period of the sampling season in which the individuals were tagged. Level 3 processed products with a temporal resolution of 30 days and spatial resolution of 4 km were used, captured by the AQUA MODIS (Moderate Resolution Imaging Spectroradiometer) sensor installed onboard the EOS PM-1 satellite. Chlorophyll-a concentration is often used to identify areas of higher biological productivity in the ocean, and it often indicates intermediate consumers abundance (macrozooplankton and small epipelagic fish), which ultimately relates to the presence of top predators, such as seabirds (Supplementary Material Table S7). Collinearity among variables was assessed separately for each colony by calculating pairwise Pearson correlations and variance inflation factors (VIF)[44]. Correlation analyses were performed using the R packages 'corrplot' (v 0.92)[45] and 'usdm' (v 1.1–18)[46]. All oceanographic variables were not measurably collinear ($VIF \leq 5$) and were uncorrelated ($r < \pm 0.75$).

MaxEnt models were carried out for each of the colonies and time period (day or night). Subsequently, models were projected to adjacent storm-petrel colonies and weighted according to colony size (number of breeding pairs)[9] (Supplementary Material Table S8 and Fig. S7). For each model, we created 5 replicates, using 70% of the data to build the model and 30% to validate it. Finally, the 5 replicates were averaged. All the models were evaluated using a cross-validation[47]. Response curves were analysed for each of the environmental variables used in the model. Additionally, a Jackknife test was conducted to quantify the contribution of each variable individually, measured by the AUC (Area Under the Curve). Maps shown in Figs. 1 and 3 were created by the authors using QGIS Geographic Information System (version 3.40.12-Bratislava).

Flight altitude

Understanding the vertical movement dimension is key to assessing bird behaviour, nevertheless studies conducted to date have primarily focused on evaluating device accuracy in the horizontal plane, with much less emphasis on the altitudinal axis. As a result, there is a lack of accurate information on flight altitudes and their associated errors. Devices that rely on GPS triangulation to estimate altitude (like Pathtrack devices) typically overestimate altitude during horizontal movements and underestimate amplitude during vertical movements, which should be considered when analysing the flight heights of individuals tagged with GPS devices[48].

To correct the altitude recorded by the GPS devices deployed on storm-petrels, altitude data were processed as follows: (1) negative values were retained[30] and extreme values were removed by excluding the upper and lower 2.5% of observations and; (2) locations obtained from georeferenced fixes triangulated with seven or

fewer satellites were discarded, thereby reducing altitude error ~ 9 (mean absolute error), i.e. altitude values should be interpreted within approximately ± 9 m uncertainty; (3) Ellipsoidal heights (WGS84) were converted to orthometric heights (mean sea level) by subtracting the EGM2008 geoid undulation at each location[49]; and (4) the bias estimated for the PathTrack-derived altitude was subsequently subtracted from the geoid-corrected height. To estimate the bias in altitude recorded by the PathTrack devices, we calculated height above mean sea level using "resting" fixes, applying the strictest criteria possible by selecting only positions with calculated speed < 1 km h^{-1}, and retaining only cases with at least two consecutive fixes to ensure true resting on the sea surface. This subset comprised 76 fixes (mean = – 5.76 m; median = – 5.55 m; IQR = 18.05 m), and the corresponding central tendency was used as the device-specific altitude bias for the final correction.

The analysis of flight heights has been conducted separately for daytime and nighttime data. The analysis of flight heights was conducted using RStudio version 2021.09.0 Build 351 (Ghost Orchid Release, 2021-09-20) for Windows.

Analytical procedures

Differences in trip duration, total distance, maximum distance travelled, and mean and maximum speed of monitored European storm petrels among colonies were evaluated using a one-way analysis of variance (ANOVA). Likewise, we evaluated flight height using a two-way ANOVA with an interaction term, testing the effects of time of day (day vs. night) and bird behaviour (resting or feeding vs. movements). When significant effects were detected, post hoc comparisons were conducted using Tukey's HSD test. Analyses were performed in R, with statistical significance set at $p < 0.05$.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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Author contributions

All authors conceived the study and collected the data in the field (M.Y, J.Z, M.G, M.I, I.A, S.R, L.Z, J.B, I.Z, C.M). J.B., I.Z., and C.M. analysed the data. C.M., I.Z., and L.Z. wrote the main manuscript text with contributions from all co-authors. C.M. and J.B. prepared Figs. 1–4 in the main manuscript and Figs. S1–S7 in the supplementary material. All authors reviewed the manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

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Correspondence and requests for materials should be addressed to C.M.

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